

ROADSIDE

Your vehicle's roadside mechanic.

Hold your phone near your vehicle. It listens, looks, and tells you what to do — fully on-device.



THE PROBLEM

A new noise, and only two bad options

IGNORE IT

Keep riding and hope.

A worn part fails on the road, and a small engine problem becomes an expensive one.

GO TO A MECHANIC

Travel, wait and pay — just to find out.

Often far away, often for something you could have checked yourself in five minutes.

There's nothing in between.

THE GAP

Drivers can describe the sound. Not the fault.

"It rattles when I ride"

"It knocks when I accelerate"

"It squeals when I brake"

Spark plug?

Brake pads?

Drive belt?

Chain & sprocket?

Everyone can hear this.

Nobody knows what it means.

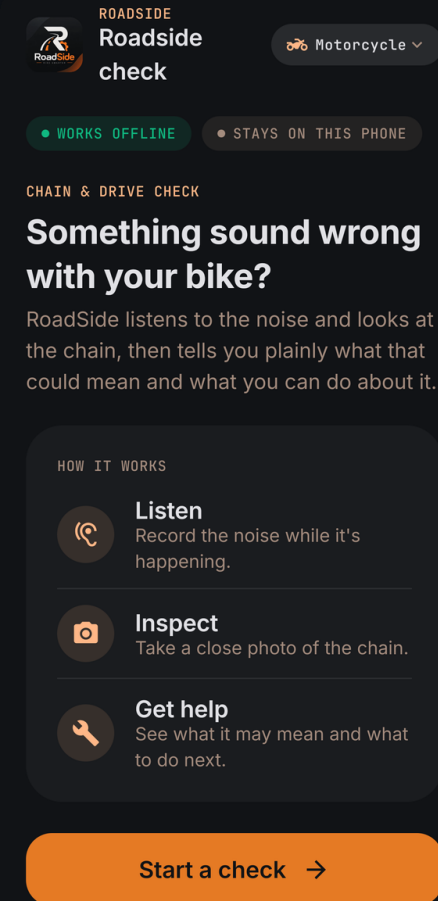
THE IDEA

Your phone becomes a roadside mechanic

Record the noise. Photograph the part.

RoadSide listens to the sound, looks at the photo, and tells you in plain words what it found, what it may mean, and what you can do — with a safety note and when to see a mechanic.

No internet. No account. Nothing leaves the phone.



WHAT ROADSIDE CATCHES

Current + planned checks, across bikes and cars

• Working today ○ Planned

• Chain rattle	→	Clean and lube the chain, check slack	BIKE
• Worn sprocket teeth	→	Ride slowly on smooth roads, see a mechanic	BIKE
○ Brake squeal	→	Check the pads, never oil near the brakes	BIKE + CAR
○ Starter clicking	→	Check battery terminals, charge the battery	BIKE + CAR
○ Engine knocking	→	Ease off, check fuel and engine oil	BIKE + CAR
○ Misfire / spark plug	→	Inspect or replace the spark plug	BIKE + CAR
○ Belt squeal	→	Check belt tension and wear	CAR
○ Uneven idle	→	Check air filter and plugs, then a mechanic	BIKE + CAR

HOW IT WORKS

Three stages, all on the phone

1

Listen

An on-device sound model picks out the fault in a short recording — a rattle, a knock, a squeal or a misfire.

2

Look

An on-device vision model checks the photo for visible problems — a dirty chain, worn teeth, a corroded terminal.

3

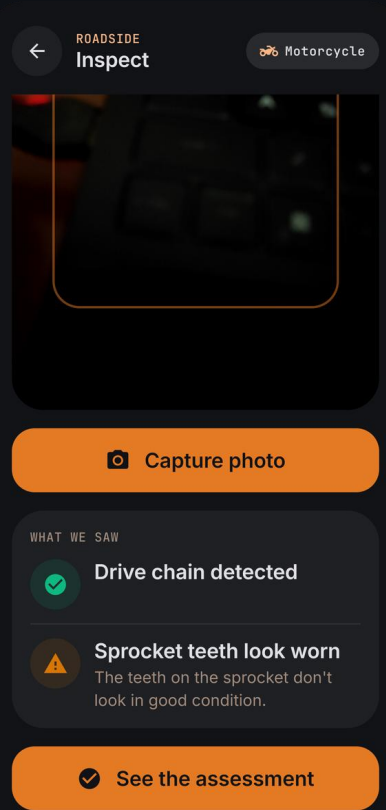
Advise

A rules engine combines sound and sight into one assessment, a plain-language solution and Ask RoadSide.

Only the microphone and camera decide. What the rider types never changes the result.

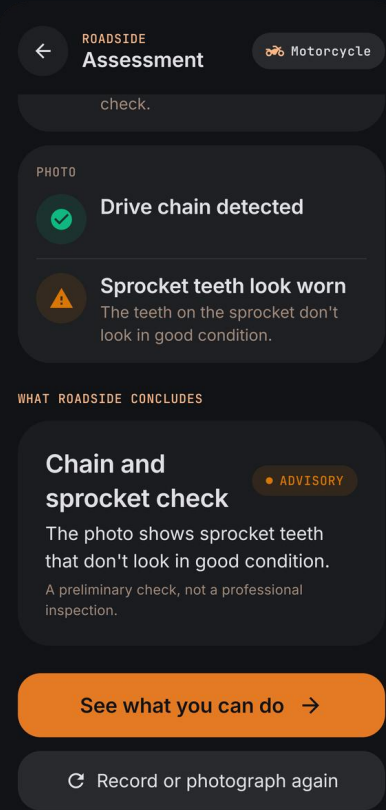
SEE IT WORK

From one photo to plain advice



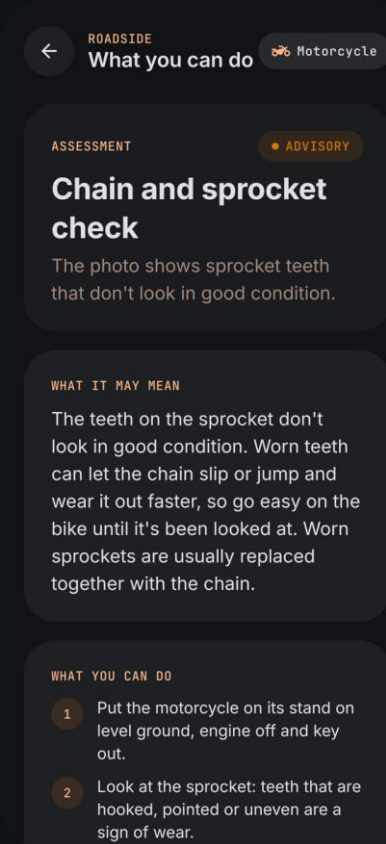
Inspect

Chain detected, sprocket teeth look worn



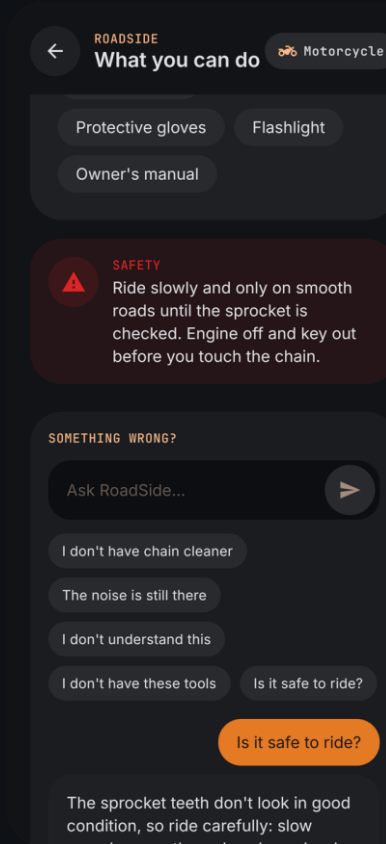
Assessment

What it saw, kept apart from what it concludes



What you can do

Meaning, actions, tools and safety



Ask RoadSide

"Is it safe to ride?" answered in context

THE PROTOTYPE

What we've already built

We built the first check end to end — chain and drivetrain — to prove the whole pipeline runs on a phone.

Sound

YAMNet — an open, on-device sound model — with our trained drivetrain-noise head

Vision

MobileNetV3 — an open, on-device image model — with our trained head for the chain, visible dirt and possible sprocket wear

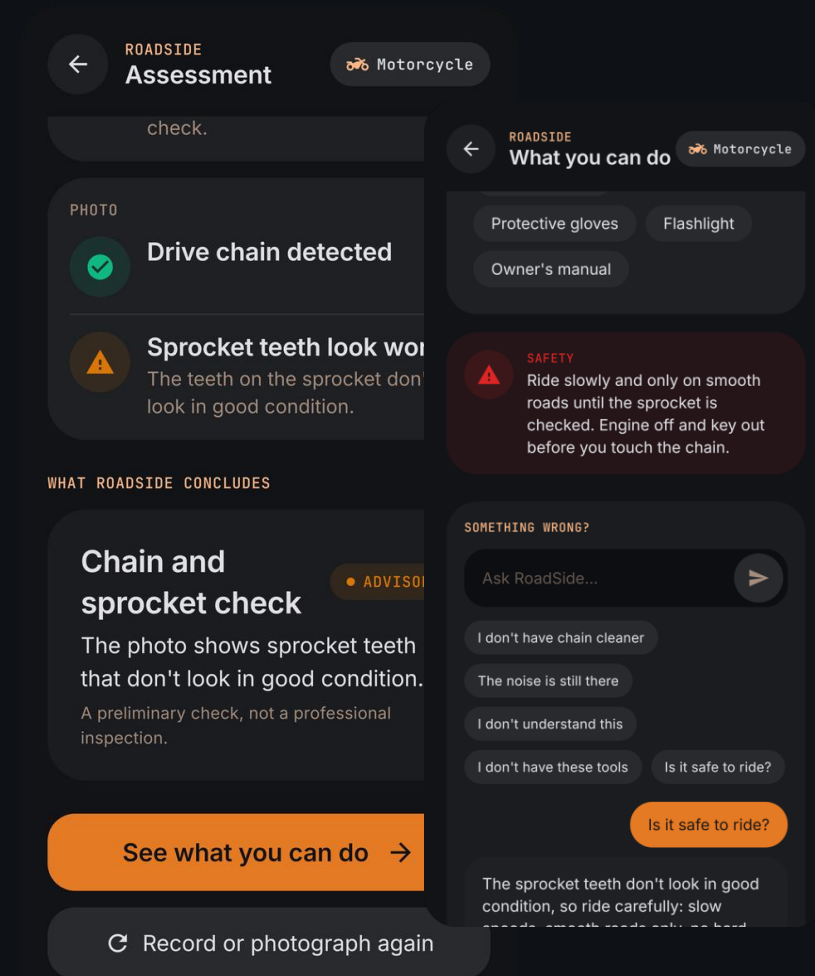
Advice

Rules engine that fuses sound and sight into a plain-language solution, plus Ask RoadSide

Runs on

Android · Kotlin · Jetpack Compose · CameraX · LiteRT — fully offline, no internet permission

At the hackathon, the same pipeline grows to engine, brake and car checks — one small trained head per fault.



WHY ON-DEVICE

It works where breakdowns happen

Instant

No upload, no server queue. The answer appears right after you record.

Private

Recordings and photos never leave the phone.

Anywhere

Airplane mode, no signal, no data pack — still works.

Free to run

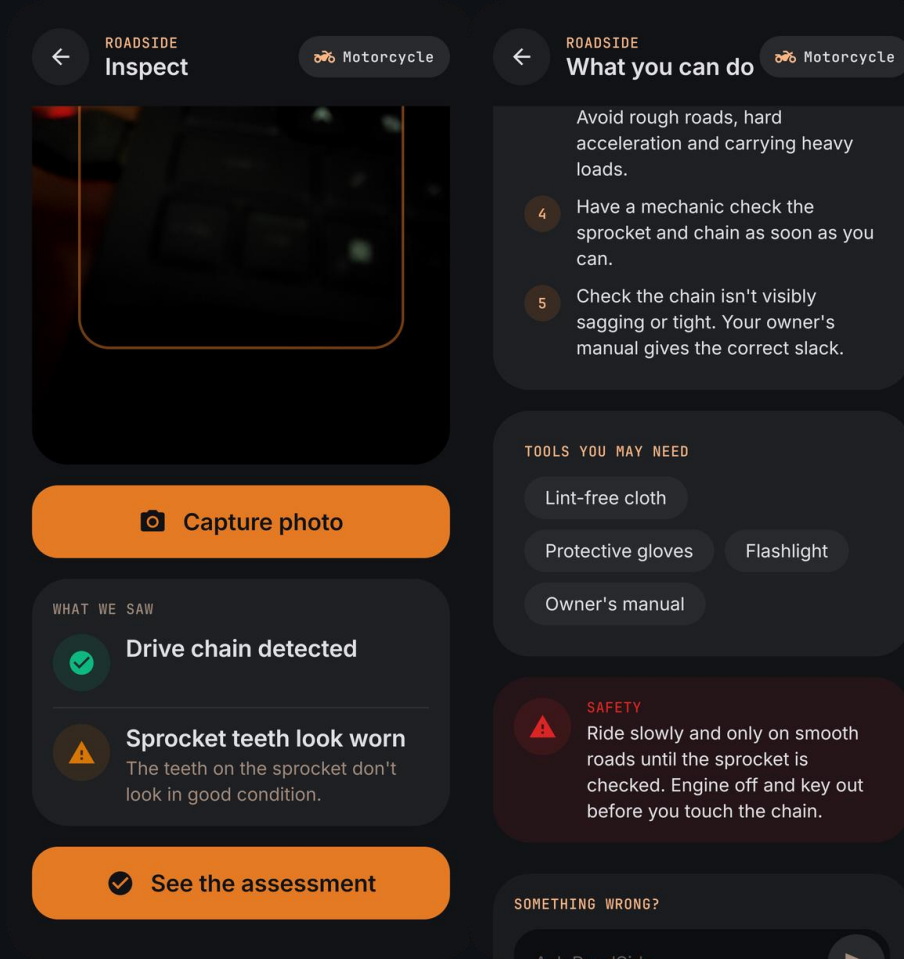
No per-use cost, nothing to host.

Compact sound and vision models run on the phone itself, with no internet permission at all.

THE DEMO

Live on the phone, in airplane mode

- 1 Record the noise from the vehicle
- 2 Photograph the part
- 3 RoadSide shows what it heard and saw
- 4 The assessment and what to do next
- 5 A judge asks “Is it safe to ride?”



What we're claiming, and what we're not

A first check, not a mechanic

RoadSide tells you what to look at and when to get help. It does not replace an inspection.

Unsure means unsure

When a sound or photo isn't clear, it says so and asks for a retake instead of guessing.

No invented measurements

No fake frequencies, millimetres or percentages — only what it actually observed.

Built to be the first thing a rider opens, and honest about when to see a mechanic.

Stack and team

Stack

- Android, Kotlin, Jetpack Compose, CameraX
- Compact on-device sound model with a small trained head per fault
- Compact on-device vision model for visible wear and damage
- On-device rules engine and Ask RoadSide
- No network permission, no cloud services

Team

D Sri Sashank

M N B Prasad Reddy

L Atchut Kumar

A new noise means guesswork.

A mechanic means a trip.

RoadSide tells you right there.